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Exploring paradoxical tensions in scaling circular business models within the textile and clothing industry☆

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ABSTRACT

The Textile and Clothing (T&C) industry is increasingly adopting circular business models (CBMs) to promote sustainable production and consumption. However, scaling CBMs within this sector remains challenging, mainly due to conflicts in organizational objectives, product designs, and value chain structures, resulting in paradoxical tensions (PTs). The purpose of this study is to explore PTs that hinder the scaling of CBMs in the T&C industry. The methodology adopted a two-stage approach: first, through a critical literature review, seven key PT categories affecting CBM scaling were identified, and secondly, semi-structured interviews were conducted with T&C industry actors for in-depth exploration. By applying paradox theory, this study revealed that PTs are more prevalent at the product and process levels. At the product level, PTs impacted scaling CBMs both within inner loops (repair and reuse) and outer loops (recycling), driven by socio-environmental and economic conflicts, classified as performing paradoxes. Scaling recycling is hindered due to trade-offs in technology selection for higher market uptake, lack of infrastructure, and challenges in choosing between local and global value chains, classified as organizing PTs. Our findings emphasize the importance of inter-class organizing-performing paradoxes, which obstruct the scaling of sequential loops like reuse-recycling. Furthermore, implicit belonging and learning PTs became apparent alongside other paradoxes, such as performing-belonging and organizing-learning. While belonging PTs arose at the stakeholder level due to identity tensions, learning PTs stemmed from insufficient data, knowledge, and technology. Based on the findings, this study contributed to developing four propositions explaining the impact of PTs in scaling T&C CBMs. These findings provide valuable insights for T&C actors to strategically plan and scale CBMs while understanding and navigating PTs. This is of crucial significance for policymakers in formulating effective policies supporting the scaling of CBMs within the T&C industry.

1. Introduction

Circular Economy (CE) has emerged as an alternative economic paradigm for achieving sustainable production and consumption targets, addressing the UN's Sustainable Development Goal 12, which promotes efficient and responsible use of shared natural resources (Dagilienė and Varaniūtė, 2023; De Angelis, 2021). As a disruptor of the linear industrial economy, CE – and embedded circular business models (CBMs) – not only reduce resource use and waste generation through circular product/material loops, but also provide a wider systemic logic that creates, captures, and delivers value for a broader group of stakeholders (Dagilienė and Varaniūtė, 2023). To maximize this value potential and foster economic, social, and environmental benefits, scaling CBMs is essential, as it drives system-level changes that go beyond revenue generation, such as shaping new regulations and driving societal transformation (Han et al., 2023; Klofsten et al., 2024). As stated by the European Environment Agency (2019a), *"to achieve real impact beyond mere optimization of the status-quo, CBMs need to scale and achieve significant market penetration"* (p. 29). In this context, the notion of scaling can be drawn from Bocken et al. (2018), describing it as the final

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phase in CBM innovation process. Being the "final" phase of the process of CBM implementation, scaling necessitates continuous organizational transformation, driven by the preceding phases of iterative experimentation, piloting, debriefing, and learning (ibid.). This implies that without scaling, the CBM implementation process would not be fully realized.

A central premise of this paper is that, in context to CBMs, although scaling is embedded within the process of implementation (Hultberg and Pal, 2021, 2023), fundamentally it involves more complex organizational set-up including new capabilities, work procedures, organizational relationships, and technologies (Lopes de Sousa et al., 2019; Sandberg and Hultberg, 2021). A reasonable conjecture is that such set-up is also confronted by complex organizational challenges (De Angelis, 2021). Typically, these complex challenges are resultants of divergent inter- and intra-organizational operations and processes, changes in production models, and contrasting organizational logics, as well as routines and strategies that have evolved during the iterative phases of experimentation, testing or piloting of CBM implementation – further complicating scaling efforts (Bocken et al., 2018; De Angelis et al., 2018). Additionally, with differing values at stake, sometimes competing in nature, the scaling of CBM initiatives can yield paradoxical tensions that are inherently intertwined and stem out from different CBM practices. Hofmann and Jaeger-Erben (2020, p. 2784) thus postulate that *"organizational transition management of CBM innovations requires the management of paradoxes"*, due to the presence of contrasting value creation systems. Paradoxes or paradoxical tensions (PTs), defined as *"contradictory yet interrelated elements that exist simultaneously and persist over time"* (Smith and Lewis, 2011, p. 382) arise as organizations transition through different phases of development, resulting in contradictory yet interrelated demands among stakeholders' goals, decisions, and value chain operations (Dagilienė and Varaniūtė, 2023; Smith and Lewis, 2011). Therefore, it is crucial to understand how fostering scaling is accompanied by the emergence of PTs, especially when systemic contradictions are on a larger scale as environments become increasingly global, dynamic, and competitive (Smith and Lewis, 2011).

These PTs, for which empirical research in the context of CBM scaling is limited (De Angelis, 2021; Hopkinson et al., 2018), can lead to unintended consequences that threaten systemic circularity. Acknowledging such PTs in scaling CBMs is crucial, as it aids in tackling the complexities of sustainability (van Bommel, 2018). As emphasized by Smith and Lewis (2011), PTs must be managed rather than simplified into binary trade-offs or dilemmas, or viewed merely as barriers and challenges. Despite the need for more nuanced research on PTs in CBM scaling, the existing literature (e.g., Daddi et al., 2019; Dagilienė and Varaniūtė, 2023) fails to provide a distinctive investigation of the topic, and instead, looks into paradoxes in the overall CBM implementation process. While current research acknowledges that scaling is significantly influenced by PTs (De Angelis, 2021; Hopkinson et al., 2018), it does not distinguish which tensions are specific to scaling. This lack of knowledge leaves a critical gap in understanding the unique PTs that emerge during the scaling phase of CBMs.

Our study is empirically anchored in the textile and clothing (T&C) industry, one of the most polluting industries in the world, involving significant material and product consumption (Salmi and Kaipia, 2022; Sandin and Peters, 2018). To minimize environmental impacts, the T&C industry is shifting towards CE, with an increasing adoption of CBMs. However, the industry faces major challenges in scaling CBMs due to rapid production cycles, fast-fashion trends, and a fragmented global supply chain (European Environment Agency, 2019a, 2019b). These factors contradict circularity principles such as durability, reuse, and recycling, limiting scaling and the ability to disrupt the existing system (ibid.). Currently, only 20 % of textile waste is reused or recycled, with the rest incinerated or landfilled (Wójcik-Karpacz et al., 2023). Various CBMs aim to address these issues by slowing down or closing resource loops, thereby prolonging product life, maximizing resource utility, and minimizing waste generation (Salmi and Kaipia, 2022). Examples include repair, take-back schemes for resale, remaking, and recycling (Ellen MacArthur Foundation, 2021), alternative consumption models like clothing libraries (Zamani et al., 2017), and garment swapping (Khitous et al., 2022), as well as upcycling and redesign (James and Kent, 2019; Paras et al., 2023). With diverse CBMs being operational in the T&C industry, the setting is empirically rich in exploring PTs that hinder current CBM scaling. One recurring PT in the literature that can be spotted is the tension between achieving product durability vs. design for recyclability. The fashion industry values fiber blends to enhance product durability and functionality (Sandvik and Stubbs, 2019), as different CBMs, such as repair and resale, aim to extend product life (Pal et al., 2019). However, such blended material composition tends to reduce ease of recycling, due to the lack of available technologies capable of producing high-quality materials that meet customer expectations (Daddi et al., 2019). Given such PTs, along with many others, and their significant impact, the purpose of this paper is *to explore paradoxical tensions that emerge during the scaling of circular business models in the textile and clothing industry*. This research gap is particularly prominent in this sector, where the thorough investigation of these tensions is crucial for effectively scaling CBMs and advancing sustainable practices.

The paper is structured as follows: Section two introduces paradox theory and its classification, followed by a critical review of PTs in scaling CBMs. Section three covers research methods, including data collection and analysis. Paradox categories, subsets, and classifications are presented in section four and discussed level-wise in section five. Finally, section six concludes the study by outlining its contributions and providing suggestions for future research.

2. Literature review

This section introduces paradox theory and its classes followed by a critical literature review that identifies PTs in CBM scaling.

2.1. Paradox theory and its classes

Building on the typology of organizational paradoxes, Smith and Lewis (2011, p. 382) define a paradox *"as contradictory yet interrelated elements that exist simultaneously and persist over time"*. This definition highlights two major components of a paradox: elements that, while inconsistent, seem logical individually (Smith and Lewis, 2011). For instance, organizational managers need to continuously improve existing operations to sustain their competitive edge while also fostering innovation to adapt to changing market demands (Schad et al., 2016). This underscores the significance of interrelatedness, emphasizing the need to balance opposing elements of a paradox over time rather than attempting to eliminate them (Smith and Lewis, 2011).

Smith and Lewis (2011) grouped paradoxes into four classes: Organizing, performing, belonging, and learning. Organizing paradoxes arise from internal dynamics, and competing organizational designs, processes, structures, and cultures (ibid.). In essence, these paradoxes arise when leaders delineate the boundaries of their operational practices, specifying *how* they are (or are not) operating (ibid.). For example, production can be subjected to contradictory elements such as the need for control vs. flexibility, centralization vs. decentralization, routine vs. change, collaboration vs. competition, efficiency vs. resiliency/creativity as well as separation vs. integration (De Angelis, 2021; Smith and Lewis, 2011).

Performing paradoxes arise from conflicting demands of external and internal stakeholders on values, strategies, and goals (Schad et al., 2016; Smith and Lewis, 2011). These paradoxes become evident when an environmental or social focus contradicts a financial focus, leading organizational leaders to differentiate between their goals and *what* they aim to avoid (Smith and Lewis, 2011). For instance, environmental sustainability efforts can increase costs, which organizations may choose not to pursue as it reduces short-term profits (De Angelis, 2021). These

paradoxes also emerge in decisions between global and local operations, as well as in balancing network-level value with company-specific values (De Angelis, 2021; Smith and Lewis, 2011).

Belonging paradoxes involve identity tensions between individuals and groups, such as independent organizational identity and organization as a part of a wider system or gaps between personal and organizational socio-environmental perspectives (Hahn et al., 2018; Smith and Lewis, 2011). Such paradoxes are thus linked to the emotional and behavioral reactions of actors when choosing between cooperation and competition (Smith and Lewis, 2011). Specifically, determining *who* will do what emphasizes these conflicting roles and values within organizations (ibid.).

Learning paradoxes arise when organizations balance the pursuit of new knowledge and innovative solutions (exploration) with the refinement of existing practices and maximization of efficiency (exploitation), or when they aim for both incremental and radical innovations (De Angelis, 2021; Smith and Lewis, 2011). Since acquiring knowledge is a multi-level construct, learning paradoxes can encompass other paradox classes (Jarzabkowski et al., 2013). Such cross-class paradoxes fall under Smith and Lewis's (2011, p. 383) analysis of six inter-class paradox types that arise from intertwined contradictions. For example, learning-organizing paradoxes occur in organizations that pursue both efficiency and agility, while performing-belonging paradoxes happen when individuals struggle to balance their personal identity with social and work demands (Smith and Lewis, 2011).

2.2. PTs in CBM scaling

This section critically reviews existing literature on PTs in CBM scaling. Due to the limited literature distinguishing CBM scaling from implementation, our review takes a broader perspective by considering the general body of literature on PTs that hinder CBM operations. This review resulted in the classification of PTs into seven categories across four distinct levels (see Table 1).

A commonly observed PT in organizations working with different CBMs arises when contrasting CBM logics for slowing, closing, or narrowing the CE loops are applied. The influence of such inherent clashes in CBM logic may intensify during scaling, as it requires dividing resource allocation between contrasting CBMs. Some authors (e.g., Charef et al., 2021a; Kühl et al., 2022) underline the inherent clash between slowing and closing logics, which is unavoidable when creating durable and functional products with synthetic materials vs. ensuring their recyclability at the end of their life cycle. Kühl et al. (2022) highlight that even though both are desired and must be scaled to meet respective objectives, durability and functionality can inhibit resource cycling, thereby hindering the development of an economically viable recycling system. A similar tension can be apprehended during the product development stage, specifically between narrowing the loop vs. closing the loop. van Bruggen et al. (2022) examine how achieving a narrowing logic in the product design stage – such as replacing metals with plastics in vehicles to reduce overall weight and, in turn, lower energy consumption – can hinder recycling efforts, as the technology for recycling plastics remains underdeveloped.

At the product level, other PTs are also reported to arise when making substitution choices (Daddi et al., 2019; Mittal et al., 2024), which involves compromising one product feature to prioritize another. Daddi et al. (2019) scrutinize the T&C industry, presenting evidence of a PT that emerges when incorporating recycled wool into new products. While this can enhance companies' environmental commitment, it may also raise health and safety concerns beyond quality issues, thereby hindering societal benefits. Such contradictions can risk brands losing their competitive advantage if customers or the market reject them, thus hindering the use of recycled materials. Mittal et al. (2024) further reveal a similar tension in the realm of footwear, where hygiene-related concerns can hinder the promotion of alternative consumption scenarios like rental and swapping.

Table 1
A deductive framework for categorizing PTs in CBM scaling.

Levels	PT categories	Contextualization in CBM scaling	Derived from
Product	1-Between different circular product logics underpinning CBMs in different CE loops	PTs arise from the contradiction between prioritizing the logic of slowing down product life cycles and focusing on narrowing loops for greater efficiency, vs. fully closing the loops.	Arnold et al. (2023); Charef et al. (2021a); Kühl et al. (2022); Prendeville et al. (2017); Sandvik and Stubbs (2019); van Bruggen et al. (2022)
	2-Between different substitution choices	PTs occur when certain material or product features are prioritized over others based on customer preferences and market demand, undermining the successful operation of CBMs.	Bressanelli et al. (2019); Condotta and Zatta (2021); Daddi et al. (2019); Jensen et al. (2022); Mittal et al. (2024)
Process	3-Between reverse supply chain disposition choices	PTs appear in the selection process of sequential loops, leading to a particular loop being ignored due to the varying degrees of materials/ products' value degradation over time, or the risk of market cannibalization.	Greer et al. (2021); Huerta Morales (2020); Kühl et al. (2022)
	4-Between alternative product management and operations strategy	PTs arise from the simultaneous need for process standardization and the demand for adaptation, upgradeability, and customization – factors that drive circular innovation but also disrupt established processes and reduce operational efficiency.	Arnold et al. (2023); Bressanelli et al. (2019); Charef et al., 2021b); Huerta Morales (2020); Prendeville et al. (2017); Roos Lindgreen et al. (2022); Sandvik and Stubbs (2019); Yang et al. (2018)
Network	5-Between circular supply network structures and their organization	PTs occur when balancing efforts to increase flexibility and cooperation while sustaining organizational unity, or when aligning centralized-decentralized and localized-globalized value chain operations.	Bressanelli et al. (2019); Brändström et al. (2024); De Angelis (2021); Greer et al. (2021); Jaeger and Upadhyay (2020); Jensen et al. (2022); Khorshidvand et al. (2023); Tong et al. (2018); Wang et al. (2017)
Stakeholder	6-Between different levels of control at policy level	PTs arise from established regulatory mechanisms and policies, disrupting circular initiatives driven by the need to balance socio-environmental standards with commercial opportunities.	Daddi et al. (2019); Johansson (2022); Tong et al. (2018)
	7-Differential interests of internal stakeholders	PTs emerge from competing interests and mindsets among internal stakeholders, causing conflicts in	Acquier et al. (2017); Kohtamäki et al. (2020); Muñoz and Cohen (2017)

(*continued on next page*)

Table 1 (*continued*)

Levels	PT categories	Contextualization in CBM scaling	Derived from
		decision-making and value alignment within CBMs.	

Another source of PTs that might hinder CBM scaling is within underlying reverse supply chain operations, particularly those that arise between choice of different disposition routes. Huerta Morales (2020) identifies a PT that hinders the refurbishment of used electronics, such as computers and mobile phones. As the value of electronics decreases over time, they lose their appeal to customers, which makes refurbishment unprofitable despite the environmental benefits of complying with CE norms. Consequently, these products are redirected toward recycling. In a similar vein, while reuse or refurbishment is preferred over recycling according to the CE waste hierarchy, Kühl et al. (2022) emphasize the risk of developing a 'secondary' brand image and market cannibalization associated with such CBMs within the UK and German manufacturing sectors, potentially directing usable items toward recycling. Additionally, PTs have been reported to arise when organizations employ alternative strategies to achieve operational efficiency. Some authors (e.g., Prendeville et al., 2017; Sandvik and Stubbs, 2019) discuss the use of innovative bio/recyclable materials in closed-loop systems and the challenges of implementing them due to logistical inefficiencies. Both process standardization and innovation are essential for adopting CBMs. Furthermore, there are tensions between standardization vs. upgradability/adaptation (Charef et al., 2021b; Yang et al., 2018) or standardization vs. customization (Bressanelli et al., 2019; Huerta Morales, 2020). For example, Huerta Morales (2020) highlights the challenge of balancing customization and aesthetic considerations with the necessity of standardization in circular product design, especially when developing reusable and upgradeable signage. This tension arises as designers strive to meet specific aesthetic goals while ensuring products can be standardized for reuse after deployment.

PTs also frequently arise when deciding between contradictory ways of organizing circular supply networks and the choice of governance options (Bressanelli et al., 2019; Jensen et al., 2022). Governance choices, like collaboration and outsourcing, are recognized for their potential to enhance circularity through shared values and resource efficiency. However, the literature points out significant tensions associated with these approaches. For example, Bressanelli et al. (2019) highlight that while collaboration fosters value co-creation, it can also raise concerns over the protection of intellectual property (IP) rights. To safeguard proprietary knowledge, manufacturers may restrict third-party access to spare parts, repair manuals, and tools, thereby limiting the scaling of CBMs. In the medical equipment sector, strong IP protections create barriers for third parties seeking service contracts for repairs and maintenance, ultimately restricting circularity. Furthermore, inadequate execution of outsourced activities can pose reputational risks for firms (ibid.).

At the network level, tensions can also arise across multiple dimensions, including disparities in material sourcing across regions and the trade-off between efficiency and resilience (Brändström et al., 2024; Khorshidvand et al., 2023). Building upon the insights from Khorshidvand et al. (2023), centralized models deliver the highest overall profitability across the chain but negatively impact greening efforts, which are essential for CE goals. In contrast, decentralized models support stronger environmental performance and more equitable distribution of responsibilities among stakeholders, yet they fall short in achieving comparable financial returns. Brändström et al. (2024) further highlight a tension in the T&C sector related to operating the value chain locally vs. globally. While locally sourced recycled materials reduce carbon emissions due to shorter transport distances, their limited availability poses challenges for scaling recycling efforts. Conversely, global sourcing addresses supply constraints but increases transport costs, environmental impacts, and product prices, potentially reducing consumer acceptance of recycled products.

Stakeholder conflicts are another key factor in generating PTs when adopting CBMs. On the one hand, regulatory-related PTs arise from the challenge of aligning policymaker priorities, particularly as regulatory frameworks become increasingly complex (Johansson, 2022; Tong et al., 2018). Lenient regulations on waste circulation generally facilitate waste valorization by allowing industries to repurpose waste as feedstock and encouraging cross-border waste trade. However, these policies can also heighten environmental concerns and health risks, especially when waste recovery is handled manually at dumpsites. Conversely, stricter regulations – like those in the EU – aim to reduce these risks but may unintentionally limit business opportunities by restricting inter-industry transactions. On the other hand, internal stakeholder dynamics can contribute to PTs, as individual perspectives shape conflicting priorities within organizations (Acquier et al., 2017; Muñoz and Cohen, 2017). For instance, the literature indicates that sharing-based CBMs are designed to scale and should focus on enhancing customer engagement and promoting social sustainability. However, financial pressures to maintain platform viability can lead to a shareholder-driven focus on economic returns, potentially weakening the collaborative efforts necessary for fully scaled CBMs (ibid.).

A critical analysis of the literature, as detailed in the methodology section, revealed seven primary categories of PTs across four levels (Table 1), which can impact CBMs and be applied to the context of scaling.

3. Methodology

Our study employs an abductive approach, beginning with a critical review of the CE literature to build a foundation for understanding PTs within the CBM framework, intending to extend it to the context of scaling. This is mainly due to: (i) insufficient literature on the CBM scaling, especially within the T&C industry, (ii) dearth of distinctive literature on PT that explicitly addresses CBM scaling, despite a large cross-section of CBM implementation papers suggesting its implicit relevance, and (iii) inherent ambiguity in distinguishing terms like dilemmas, barriers, and challenges from paradoxes due to their implicit nature (Smith and Lewis, 2011). Taking these points into consideration, a critical review in this context was deemed essential for conducting an interpretative exploration of the relevant knowledge base *"to potentially conceptualize, and to expand on the theoretical foundation"* (Snyder, 2019, p. 336) of PTs in the context of CBM scaling.

Although there is no strict standard for conducting a critical review, and this approach requires researchers to have advanced skills and to document the transparency of the process, it is particularly useful for addressing new, emerging topics (Snyder, 2019). Given that critical reviews allow for a more creative collection of data and the combination of perspectives from different fields or research traditions (ibid.), we began by searching reputable databases such as Scopus, Web of Science, ABI/INFORM Collection, ProQuest One Business, and Business Source Premier. Including the fields of business, management, and social science, we examined various industries and contexts such as T&C, furniture, automotive, food, construction, wood, among others. This broad approach was necessary because identifying key paradoxes is crucial for macro-level transformation in the T&C industry. To fully cover this context, we also expanded our search beyond the term *circular economy* to include related concepts such as the sharing economy, product-service systems, and reverse supply chains. Additionally, we examined terms that are often used interchangeably with paradoxes, such as conflicts, dilemmas, barriers, and challenges. By carefully examining 122 retrieved articles and analyzing broader semantic patterns in line with the definition of paradox by Smith and Lewis (2011), we formulated seven main categories of PTs across four different levels, based on an in-depth analysis of 22 articles. These categories, as presented in

Table 1 in section 2.2, can be extended to the context of CBM scaling. This deductive framework has been used to develop the interview guide and gather data through semi-structured interviews with T&C industry actors (as detailed in section 3.1) to delve deeper into the PTs encountered when scaling their CBMs. An inductive method of data analysis (as explained in section 3.2), guided by the seven categories, resulted in initiating the first-order concepts and then revealed 15 second-order themes, along with their PT classifications.

3.1. Sampling and data collection

A purposive sampling approach was adopted to create opportunities for comparison by avoiding repetition and to obtain information-rich data for the study (Freeman, 2005). Accordingly, 22 T&C actors (Appendix A, Table S1), including six fashion retail brands, six second-hand charities, six integrated sorters, and four recyclers, were selected after reviewing their CBM initiatives and their inherent scaling operations. The fashion retail brands operate CBMs related to renting, repairing, and reselling, but also engage with wider value chain actors for conducting remanufacturing and recycling. They also work with second-hand charities, who collect, pre-sort, and grade used clothes for resale on a relatively larger scale and promote remaking activities. Within the T&C reverse supply chain, integrated sorters also play a crucial role in conducting several activities; the integrated sorters in this study mainly engaged in the collection and sorting of used clothing into fractions suited for resale or material recovery, as well as in value-adding processes like recycling and downcycling. They often partner with fashion retail brands to scale the use of their mechanically recycled fiber, but also create value through downcycling, i.e., finding applications like industrial wipers and rags. Finally, the recyclers focus primarily on chemical and mechanical recycling to generate cellulosic fibers, textile pulp, and biodegradable raw materials, which are supplied to T&C producers to support closing the loop.

The selection of the actors was mainly tailored to the Swedish context. However, due to the lack of Swedish actors adequately representing all circular processes such as sorting and mechanical recycling, European actors were also purposefully included in the study. The selection process involved searching each organization's website and scrutinizing their CBM activities and reverse supply chain operations. This included retrieving data related to scaling these activities and initiatives. Where information related to scaling was difficult to obtain, the first round of interviews was conducted. This helped to better understand their CBM operations and scaling activities. In particular, some actors, mostly second-hand charities and integrated sorters, had limited information on CBM scaling-related activities.

Actors were contacted by email to organize 36 interviews: including 7 with fashion retail brands, 12 with second-hand charities, 12 with integrated sorters, and 5 with recyclers. Additionally, a brief overview of the study and an interview guide (Appendix A, S2) were provided beforehand to facilitate better reflection. The interview guide was designed with 'what' and 'how' questions to obtain in-depth insights into the PTs that could emerge during scaling operations. Interviews with company representatives were sometimes conducted with two respondents to ensure comprehensive answers and with three interviewers for better inquiring. All actors were exposed to seven categories of PTs to reveal latent PTs through more discussion. The interviews conducted online, in English, each lasted approximately one hour, and recorded with verbal consent and in compliance with the GDPR and the Swedish ethical guidelines on voluntary and anonymous participation. After each interview was completed, transcription was done to carefully analyze the implicit nature of the PTs and re-correspondence if other information was needed.

3.2. Data analysis

PTs related to scaling operations were extracted and analyzed after careful re-reading of interview transcripts and thematic coding of qualitative data, inspired by the approach of Gioia et al. (2012), as shown in Fig. 1. The inductive analysis focused on identifying contradictory yet interconnected elements that appeared during scaling CBMs. This process involved carefully examining conflicting issues, weighing their potential advantages and disadvantages, and understanding why both are essential for scaling and achieving CE solutions. After identifying a PT, we applied the "cut and sort" technique (Ryan and Bernard, 2016) to refine and concretize paradox elements, leading to the development of first-order concepts. Through repeated immersion in PTs, we also identified and excluded those related to scaling linear models from circular models, as fashion retail brands were involved in two distinct contexts. In the next step, we inductively differentiated the final PTs based on their sources of emergence to determine their classification into PT categories. For example, competitive processes and value chain structures were classified under the organizing class, dimensions of knowledge fell under the learning class, etc., in alignment with Smith and Lewis (2011). By developing 15 sub-themes (second-order themes) from the first-order concepts, they were mirrored in the seven categories of our critical review.

3.3. Research quality

The research quality has been ensured in various ways. First, involving diverse actors with different CBMs in this study helped prevent data duplication (Freeman, 2005) and improved clarity during interviews. Instead of directly using the term paradox during interviews, we discussed tensions related to scaling operations. In this way, we encouraged respondents to describe the various challenges they encountered, using examples from our critical review. This approach was well-received by respondents, helping them disclose the complexities in scaling operations and leading to the identification of new PTs not previously recognized in the critical literature review. For instance, an ethics-related paradox emerged and was later validated by other actors after being included in the interview guide. This helped ensure the credibility of our data by incorporating different voices (Pratt et al., 2020). Second, our critical review identified scaling-related PTs across various contexts and industries, and these findings were confirmed by T&C actors. This enhanced the study's analytical depth and supported the generalizability of the findings. Finally, in explaining the dependability and confirmability of the research, we thoroughly explored the term scaling, as respondents described it in a variety of ways, including *"high volume"*, *"outsourcing"*, and more. This comprehensive exploration allowed us to cover all aspects of the concept. Alongside this, we employed a structured approach to replicate the dataset, ensuring transparency and reliability at every stage. This process included revisiting the coding cycles when necessary to uphold inter-coder reliability (Seuring et al., 2012), and engaging in individual assessment of polished data to prevent biased judgments. This approach was implemented to maintain consistency and achieve a consensus among the authors on possible interpretations, as recommended by Alvesson (2003).

4. Results

This section provides an exploratory analysis of the PTs that arise during scaling CBMs in the T&C industry in each of the seven categories to showcase emergent sub-themes.

4.1. Between different circular product logics underpinning CBMs in different CE loops

The two primary types of PTs identified in this category are: (i) durability logic (for slowing the loop) vs. recyclability logic (for closing the loop), and (ii) product-service logic vs. fast fashion logic. Regarding the former sub-theme, Fashion retail brands #3 and #4 emphasized that

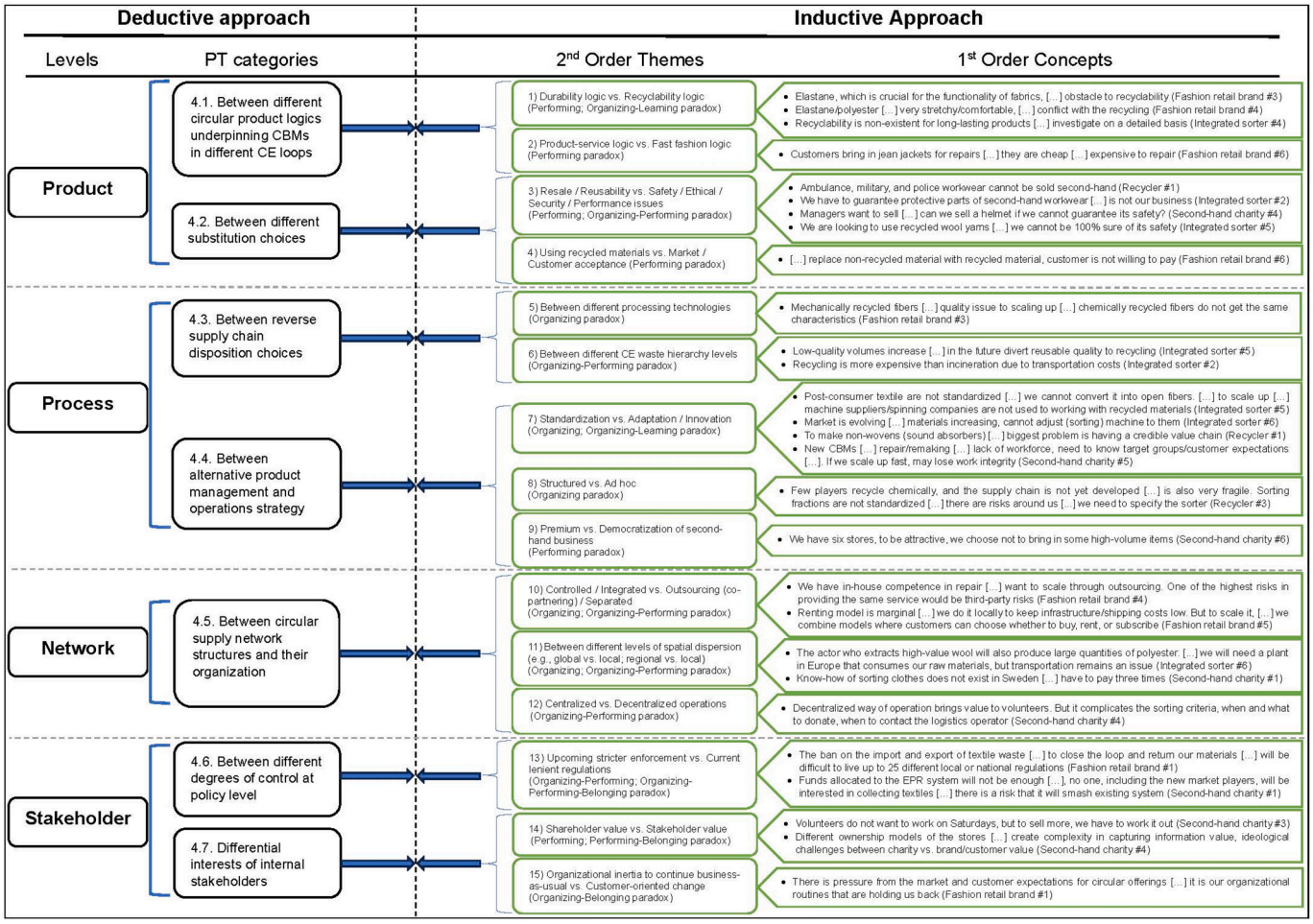


Fig. 1. Data structure.

while blends enhance product longevity and performance, they directly hinder the scaling of recycling at the product's end-of-life stage.

"We need material mixes to enhance longevity, but at the same time, they cannot be recycled. For example, elastane, which is crucial for the functionality of many fabrics, poses a significant obstacle to recyclability." (Fashion retail brand #3)

"In the case of skinny fits, a certain amount of elastane or polyester synthetic fibers is required to make them very stretchy and comfortable to wear, which may conflict with the mechanical recycling in the next step. […] I can say that this is the most typical conflict when it comes to denim." (Fashion retail brand #4)

This highlights a performing paradox between commercial (customer) value vs. environmental value. Integrated sorter #4 also underpins an organizing-learning PT, arising from insufficient data and know-how about product lifetime – such as the duration a product/material can last, the optimal time for recycling, as well as details of production processes – especially when attempting to scale recycling:

"The recyclability is non-existent now for long-lasting products. For example, jackets that are made of some synthetic materials. Thus, durability always has an impact on recycling processes. This is something we really need to investigate on a very detailed basis, maybe on a material basis, because most of the environmental cost of a product is during the production phase. […] some products are highly recyclable, although they may last only a few years, while others cannot be recycled at all but can last for a longer period." (Integrated sorter #4)

Working with inner loop CBMs, Fashion retail brand #6 noted that while repairs can slow down the loop and promote environmental sustainability, scaling of such product-service models in the fashion industry is difficult due to the high operational costs, compared to today's fast fashion trends. This underscores a performing paradox arising from conflicting values.

"From the perspective of the European Union, if it is cheaper (in fashion), then you should do it. We have a lot of customers bringing jackets that are 25 years old, and they just want to repair them. But jackets are cheap today, and it is more expensive to repair them." (Fashion retail brand #6)

4.2. Between different substitution choices

Two types of substitution choice PTs emerged within this category: First, ensuring safety and functionality while reselling certain categories of second-hand clothes. As indicated by Recycler #1, ambulance, military, and police work clothes cannot be sold second-hand due to security issues. Additionally, Integrated sorter #2 pointed out that despite the demand for second-hand workwear, such as personal protective clothing, the inability to guarantee their functionality and performance prevents them from being sold second-hand. This, in turn, hinders the scale of second-hand sales for these product categories. While highlighting a performing paradox in both instances, one of the charities elaborated on how safety and ethical concerns restrict the scaling of the resale business:

"Our store managers want to sell as much as possible, but as a children's rights organization, we have to make decisions about products that we feel are not safe to put on the market as second-hand items. These include helmets, products that may contain chemicals (e.g., old textiles from the 60s), cosmetics from a hygienic point of view, fake brands, etc. This affects our sales, but would it not be more environmentally sustainable to sell these items and

give them a second life cycle? However, can we sell a child a helmet if we cannot guarantee its safety? Or it is better for a mother who cannot afford a brand-new helmet to at least have the option to buy a second-hand?" (Second-hand charity #4)

Safety concerns also impede the uptake of recycled yarns in the new apparel production value chain. Integrated sorter #5 emphasized the importance of ensuring that recycled wool yarns are clean and free from harmful chemicals. However, achieving this goal becomes increasingly challenging when the feedstock is sourced from multiple countries. The complexity is compounded by the high costs involved in assessing every batch for contamination. Although the push for sustainability encourages the use of recycled materials, expensive procedures required to ensure their safety and quality impede effective scaling, highlighting an organizing-performing paradox.

Secondly, beyond hygiene factors, Fashion retail brand #6 notes that incorporating a high percentage of recycled materials makes products too costly and less attractive to customers from a pricing standpoint. This, in turn, could negatively impact their sales and competitiveness. This PT exemplifies a performing paradox, where the conflict between economic and environmental values poses a direct barrier to scaling use of recycled materials.

"When it comes to recycled materials, if we simply replace non-recycled material with recycled material, the customer is not willing to pay more to make up for the lost margin elsewhere. For example, take the training tights: if we use only one recycled material and charge as much as a high-end brand does, customers will not be willing to pay for it, even though recycled materials have become almost like a hygiene factor that they expect." (Fashion retail brand #6)

4.3. Between reverse supply chain disposition choices

When considering alternative disposition routes to valorize used clothing, PTs are noticeable at both, intra- and inter-process levels along the CE waste hierarchy model. For example, intra-process level PTs occur when deciding the suitability of recycling technology, as indicated by Fashion retail brand #3:

"When it comes to the availability of recycled fibers, there are mechanically recycled fibers, but quality is an issue. This is our biggest barrier to scaling up. That is why we had to work with a [chemical recycler] and use chemically recycled fibers to eliminate the quality problem. But instead, we do not get the same characteristics – for example, cotton does not look like as it should." (Fashion retail brand #3)

It can be interpreted that while more widely available mechanically recycled fibers suffer from quality issues, alternative chemical recycling can be more effective in providing higher quality. However, this process somewhat transforms the fiber into regenerated material, thus losing some original properties. These competing technology choices limit the potential for scaling market uptake, as was added by Fashion retail brand #3, creating an organizing PT.

Inter-process level disposition choice PTs are reported in different contexts. Integrated sorter #5 proposed that the volume of low-quality used clothes collected in the EU will considerably increase beyond 2025 due to the upcoming Extended Producer Responsibility (EPR). While this will subsequently raise reverse logistics costs for second-hand reselling, it could also present opportunities for recycling actors, considering an increase in non-reusable volume. A risk is that reusable items might end up in the recyclable fraction to meet economies-of-scale, regardless of their potential for reuse, which undermines the environmental benefits of extending product life. The challenge of adjusting future fractions to scale CE loops, while balancing conflicting environmental and economic priorities, creates an organizing-performing paradox. There can be conflicts between other disposition routes as well; for instance, Integrated sorter #2 emphasized that, despite recycling being the preferred option over incineration in the waste hierarchy model, high transportation costs and safety concerns often make incineration a preferred alternative:

"Today, recycling is more expensive than incineration due to transportation costs or, for example, because materials or textiles are not in a condition where they can be easily recycled. When we talk about workwear, it is not sorted, and the categories include shoes, shirts, belts, or any other company- or office-related items that are too costly to sort to use for recycling." (Integrated sorter #2)

Thus, although the CE emphasizes the importance of an environmental value-based hierarchy, high processing costs and safety concerns create significant challenges in consistently adhering to it, illustrating an organizing-performing paradox.

4.4. Between alternative product management and operations strategy

Within this category, PTs related to standardization vs. adaptation/innovation, structured vs. ad hoc, and premium vs. democratization of second-hand business were predominantly identified. In terms of standardization vs. adaptation/innovation, Integrated sorter #5 states that the variability in the quality of incoming post-consumer textile feedstock makes quality standardization challenging. Alternatively, finding new production lines and partners for producing yarn from different recycled materials creates consolidation challenges, hindering economies of scale. Similarly, Integrated sorter #6 highlights that while standardizing the automatic sorting process (and technology) can optimize output feedstock, it may not be market-oriented. On the other hand, adapting the sorting process to variations in feedstock material, composition, and source can be more market-oriented but results in higher processing costs, creating an organizing paradox.

"Post-consumer textile are not always standardized […] with every garment we process, we need to work differently, [which] affects our recycling processes. So, we cannot convert it into uniformly open fibers that are suitable for yarn production. We are working on starting a new factory to scale up […] then you also engage with machine suppliers, spinning companies, etc. These companies are not used to working with recycled materials." (Integrated sorter #5)

Recycler #1 further underscores a PT in organizing a value chain for low-quality recycled materials. While collaborating with design firms to create circular products, such as non-wovens for innovative uses like sound absorbers, scaling production is challenging due to a lack of suitable manufacturing facilities. Additionally, despite growing interest, companies are often reluctant to bear the necessary costs. As one respondent noted, *"companies are not really prepared to pay. In addition to the cost, the biggest problem is having a credible value chain and production capacity to encourage a company to invest in that"*. This paradox highlights the tension between promoting innovation in circular products and the practical difficulties of establishing a scalable, investable production system.

Another PT that organizations face relates to adopting different CBMs such as repair and remaking beyond reselling. While these CBMs, in which successful results were found, can extend clothing lifetimes more than resale, they require human intervention, such as repairing and upcycling know-how, sewing skills, and addressing workforce shortages, which are scarce in high-cost countries like Sweden. Additionally, one actor (Second-hand charity #5) noted that these new CBMs might cannibalize resale volumes if many premium-quality used clothes are diverted to repair and remake operations. Balancing new CBMs with standardized operations creates an organizing paradox, while a hidden learning paradox arises from the tension between maintaining current operations and exploring new business models to meet evolving customer expectations. The company aims to better understand its customers through target group analysis, direct engagement, and events, but limited time and resources hinder this learning process. Their shift from second-hand shops to a high-end department store demonstrated how sales channels influence consumer behavior, emphasizing the need for continuous learning: *"[…] when we moved to NK (department store), we saw a completely new customer group start buying from us"*. However, with only a few people managing multiple roles, they must prioritize

daily operations over strategic efforts, making it challenging to refine existing processes while adapting to shifting market demands.

Regarding structured vs. ad hoc, Recycler #3 points out that chemical recycling demands specific categories of feedstock, which must be facilitated by a structured sorting infrastructure. However, the absence of such a structured process leads to ad hoc sorting based on changing demands, resulting in high logistics costs that hinder economies of scale. Moreover, the lack of logistics and infrastructure to facilitate scaling chemical recycling means they *"cannot even guarantee a 5- or 10-year contract"* with other industry players. This vulnerability in the recycling value chain illustrates an organizing paradox.

At the circular retail stage, an inherent PT was related to resale product management. While the Swedish second-hand resale market is growing and there is potential to add more second-hand items in the resale assortment, this may mean lower quality items are added to the retail consignment. An inherent risk is that this could bring down the overall attractiveness of the second-hand stores, thus lowering the average price level and margins. Instead, exporting a large fraction of resaleable items is deemed more strategic to keep the resale store more attractive and premium.

"About 60% of material is sent abroad referred to as a low-grade export mix. In this material, we are aware that at least 40% could be suited for the Swedish second-hand market […] but since we only have six stores, to be attractive, we have to choose not to bring in some high-volume items, as attractiveness will help us raise prices." (Second-hand charity #6)

A significant portion of the used clothes currently exported could find a market within Sweden, thereby reducing the environmental impact associated with transportation. However, these items are often suitable only for lower pricing schemes, such as pay-per-kilo, which might diminish the appeal and exclusivity of second-hand stores. Conversely, cross-border exports fetch higher resale values for these clothes but come with a greater environmental footprint. This conflict between economic value and environmental impact, which affects the scaling of the resale business model, exemplifies a performing paradox.

4.5. Between circular supply network structures and their organization

This category captures the PTs arising due to different structures and models for organizing circular supply chains and underlying operations among the T&C industry actors, such as in terms of internalization (e.g., controlled vs. outsourced or integration vs. separation), location (e.g., global/regional vs. local), and organization (centralized vs. decentralized).

In terms of internalization, Fashion retail brand #4 explained that while outsourcing could enable scaling their repair business, transferring the tacit knowledge and skills required for repairs is difficult, making outsourcing risky in terms of maintaining service quality. This contradiction between outsourcing and a controlled in-house repair business exemplifies an organizing paradox.

"We have in-house competence in repair and sell repaired items in our second-hand store, but now we want to scale these activities through outsourcing to reach a wide range of customers. That is like a separate production, and we first need to have specific knowledge about our sorting process at home to be able to transfer it for outsourcing. […] we want to maintain good control over our second-hand products to ensure they are of high quality within their segment before outsourcing, and also in terms of online platform sales, as one of the highest risks in providing the same service would be third-party risks." (Fashion retail brand #4)

Fashion retail brand #5 further declares a PT in scaling its rental business. The rental model on its own yields only marginal profits, and although it offers environmental benefits, it operates sub-optimally due to high transportation and reverse supply chain costs (e.g., laundry) and the lack of scale in business transactions. Therefore, rental services are integrated with other CBM offerings, such as resale and subscription, to give customers the flexibility to choose their options. While such an integrated approach can optimize performance through the reduction of reverse logistics costs and generating more economies of scale, this creates a risky scenario: durable clothes suited for rental and subscription, which have the potential to fetch a higher marginal price premium compared to reselling them, may end up in resale anyway, as it is more mainstream and "popular" as a CBM. Each CBM, whether running a rental business separately or in conjunction with other CBMs, has its own set of pros and cons. When coupled with the challenge of balancing economic and environmental priorities, this results in an organizing-performing paradox.

Regarding location-specific decisions, Integrated sorter #6 highlights that large-scale recycling operations, such as chemical polyester recycling plants, are not feasible on a micro-local level due to the need for a large volume of feedstock (to process) to achieve economies of scale. However, being global means higher costs of inbound transportation of feedstock, thus leading to higher environmental footprints. Being local could perhaps work for actors sorting and segregating very specific textile waste, for example, sorting wool of a certain color followed by mechanical recycling locally. However, that creates problems in terms of transporting the remaining waste fraction to other European actors regionally or even globally, thus increasing outbound transportation footprints. Such global-local operational decisions underscore an organizing paradox. Second-hand charity #1 tackles these issues in a different way:

"I am not sure how to organize this pre-sorting facility. I am very worried about the cost levels in Sweden when it comes to sorting. Based on our own experience, the know-how of sorting clothes does not exist in Sweden. We do not have enough people who know how to sort. Most importantly, they do not know how much needs to be sorted for it to be economically viable. If you compare Sweden to, for example, the Baltic countries, we have to pay three times as much, and the people over there have the culture, the productivity, and know what they have to do. But then we have to transport the low-value textiles there." (Second-hand charity #1)

This illustrates an organizing-performing paradox: scaling pre-sorting operations for used clothes within Sweden leads to high processing costs, whereas a regional set-up (such as transporting them to the Baltics) could be more cost-effective but increases the environmental footprint.

Organizing reverse supply chain operations in diverse ways – centralized or decentralized – also unfolds PTs. In used clothes handling, centralized operations enable economies of scale, as they can handle large volumes more cost-efficiently. However, decentralized operations render greater social benefits, particularly in terms of work integration and well-being, as illustrated below. This reveals an organizing-performing paradox, where the choice between economic efficiency and social value creates inherent tensions in decision-making.

"The current decentralized way of operation brings significant value to volunteers by providing work opportunities. But it complicates the sorting criteria, when and what to donate, when to contact the logistics operator for pick-up to deal with the large surplus volume of collections (which cannot be handled locally and needs more efficient ways such as processing in hubs or large production centers) […], and lack of internal visibility and consolidated know-how within the organization." (Second-hand charity #4)

4.6. Between different degrees of control at policy level

External factors, such as regulatory mechanisms, strongly influence the shaping and scaling of CBMs. PTs emerge based on the design of these mechanisms. Fashion retail brand #1 highlights that waste export-import regulations and national laws (like REACH, the Waste Shipment Directive, etc.) play a crucial role in determining the extent to which textile materials can be reclaimed and recycled to close the resource loop. Lenient cross-border regulations on waste trading can enhance recycling opportunities and scale up the market. However, transporting post-consumer textile waste from the Global North to the Global South significantly increases transportation and environmental footprints, while waste dumping remains a concern. Conversely, stricter enforcement of textile waste trade aims to prevent illegal dumping and reduce environmental impact but limits the scaling of recycling CBMs. While these regulations can improve local waste management, they pose significant challenges for international companies, which must navigate a complex array of national policies along the entire value chain. This situation exemplifies an organizing-performing paradox.

A second-hand charity explained the complexities associated with the upcoming EPR regulation, which could potentially interfere with waste management efforts:

"The funds allocated to the EPR system will not be enough to cover the costs for collecting, which will likely force you to discontinue textile collection in the near future. […] this contradicts the EU strategy, which advocates for increasing the volume of collection […] what the government must realize is that without economic incentives, no one, including the new market players, will be interested in collecting textiles. In that case, all the responsibility will fall back on the municipalities. […] With this new EU legislation and the possible EPR system within it, there is a risk that it will smash the existing system, which is currently functioning well." (Second-hand charity #1)

Such mechanisms can hinder scaling efforts across all CBMs. The complexity of designing waste collection systems under the EPR-led framework reflects an organizing paradox. This is further accompanied by an inherent belonging paradox, as unmotivated actors may prioritize their individual interests over the broader circular ecosystem (i.e., they may reject participating in waste collection). Additionally, the contradictory environmental values from increased waste collection vs. the possible economic losses due to a lack of financial incentives represent a performing paradox.

4.7. Differential interests of internal stakeholders

Two types of PTs appeared within this category. First, a clash between shareholder and stakeholder values is evident in a resale business context. Second-hand charity #3 reveals a performing paradox by noting that while scaling their resale business can boost customer value through improved services, it may diminish volunteers' motivations, who are driven by the charity's social norms. Similar inherent value conflicts, combined with dual organizational identities, are highlighted by Second-hand charity #4:

"Due to the volunteer-based engagement, the know-how level varies among retail managers. […] different ownership models of the stores (50% owned, 50% in cooperation with a local church) create complexity in capturing information value, ideological challenges between charity vs. brand/customer value, and difficulties in harmonizing the message across all stores. […] it is easier to establish new partnerships, capture information, and make value-based decisions for our stores that are owned or run by us." (Second-hand charity #4)

For internal stakeholders (i.e., employees and volunteers) working in a second-hand charity, unlike shareholders, sales may not be the primary objective. These competing roles and memberships, along with internal value-based conflicts, represent a performing-belonging paradox.

Secondly, it was evident that the status quo of linear business operations often generates enough inertia to resist the transition to a CE and the scaling efforts across various CBMs, as noted by Fashion retail brand #1. The competition between linear and circular processes reflects an organizing paradox, as linearity complicates the adoption of circular initiatives. Additionally, the reliance on linear routines to render shareholder value (via economic returns) instead of adopting a customer value proposition through circular offerings echoes a belonging paradox.

"There is pressure from the market and customer expectations for circular offerings, and our organization is a bit too slow to adapt quickly to this new demand. We are used to working with large volumes in a linear fashion, as all our processes are built around that. Therefore, it is our organizational routines that are holding us back." (Fashion retail brand #1)

5. Discussion

This study demonstrates the PTs that have a significant impact on the scaling CBMs in the T&C industry and across different levels. Table 2, derived from section 4, offers an interpretation analysis of these PTs, which arise across four levels: product, process, network, and stakeholder. Drawing upon the results, it also outlines the corresponding PT sub-themes and the specific CBM in which scaling is hindered. As shown, different PTs hinder scaling CBMs at various stages, regardless of actors' position in up- or downstream. These PTs, confirming seven main PT categories found through our critical review, occur in organizational settings due to mismatched processes, conflicting objectives, gaps in knowledge, and competing roles and identities. Their complex and implicit nature becomes particularly salient under environmental conditions and resource scarcity, highlighting the need for adaptive strategies (Smith and Lewis, 2011).

As illustrated, PTs at the product and process levels are more prominent than those at the network and stakeholder levels. De Angelis (2021) points out that tensions at the process level, such as balancing efficiency with resilience or competition with collaboration, have a direct impact on the scaling of CBMs. Similarly, product-level challenges, like incorporating secondary raw materials without sacrificing quality standards or cost-effectiveness, create strategic PTs for companies transitioning from linear to circular systems (Daddi et al., 2019). Network-level tensions, by contrast, frequently stem from process-level limitations, as firms encounter internal inefficiencies and struggle with building the necessary logistics infrastructure when transitioning to full-scale commercialization (Reim et al., 2019). Stakeholder-level tensions follow a similar pattern, as firms engage in socio-economic development only when driven by stakeholder influence (Hofmann and Jaeger-Erben, 2020).

Table 2 further reveals tensions that hinder the scaling of sequential loops, such as reuse vs. recycling or recycling vs. incineration. These PTs arise as organizations grapple with balancing operational efficiency and environmental sustainability, often disrupting the coordination of cascading operations (Huerta Morales, 2020; Kühl et al., 2022). This underscores the complexity of managing multiple loops within circular

Table 2
Impact of PTs on CBM scaling across four levels and paradox classification.

Level	Sub-theme	PT class	Type of CBM	How scaling is hindered	Supporting interviews
Product	1	Performing Organizing-Learning	Recycling Recycling	Limited market uptake due to technology limitations and circular product data	Fashion retail brand #3; #4 Integrated sorter #4
	2	Performing	Repair	Relatively expensive repair services compared to cheaper fast fashion	Fashion retail brand #6
	3	Performing	Reuse	Limited sales and growth due to concerns related to meeting safety, ethical, functionality and performance issues	Recycler #1; Integrated sorter #2; Second-hand charity #4
		Organizing-Performing	Recycling	Limited market uptake due to contamination concerns	Integrated sorter #5
	4	Performing	Recycling	Limited customer acceptance due to higher price points from higher processing and quality management costs	Fashion retail brand #6
Process	5	Organizing	Recycling	Limited market uptake due to poor output quality	Fashion retail brand #3
	6	Organizing-Performing	Reuse-Recycling	Volume-based competition and cannibalization of the share of used clothes between competing value recovery options	Integrated sorter #5
		Organizing-Performing	Recycling-Incineration	High costs (when handling textile waste) divert feedstock to lower-value recovery options	Integrated sorter #2
	7	Organizing	Recycling & Sorting for Recycling	Handling feedstock variability limits quality standardization and economies of scale; lack of credible value chains for producing innovative products	Integrated sorter #5; #6; Recycler #1
		Organizing-Learning	Repair & Remaking	High process costs, knowledge gaps, and risks of cannibalizing mainstream CBMs	Second-hand charity #5
	8	Organizing	Recycling	Lack of sorting infrastructure and high logistics costs	Recycler #3
	9	Performing	Reuse	Reduced average price point and attractiveness of high-volume, low-quality items (in second-hand consignment)	Second-hand charity #6
Network	10	Organizing	Repair	Limited customer base expansion due to operating risks related to ensuring service quality	Fashion retail brand #4
		Organizing-Performing	Rental	Volume-based competition and cannibalization of share of used clothes between competing CBMs	Fashion retail brand #5
	11	Organizing	Recycling	Lack of economies of scale and high outbound logistics costs (for waste export) in a local setup, while high inbound logistics costs (for waste import) in a global setup	Integrated sorter #6
		Organizing-Performing	Pre-Sorting for Reuse	Lack of economies of scale and availability of a skilled workforce locally, while facing high logistics costs globally	Second-hand charity #1
	12	Organizing-Performing	Reuse	Struggle to achieve cost efficiency and economies of scale without compromising social value creation	Second-hand charity #4
Stakeholder	13	Organizing-Performing	Recycling	Struggle to achieve cost-effectiveness and accountability simultaneously in the value recovery process due to a regulatory conundrum	Fashion retail brand #1
		Organizing-Performing-Belonging	Across all CBMs		Second-hand charity #1
	14	Performing	Reuse	Conflicting values, ownership models, and ideological differences	Second-hand charity #3
		Performing-Belonging	Reuse	creating misaligned CE goals	Second-hand charity #4
	15	Organizing-Belonging	Across all CBMs	Cognitive challenges to shift to circularity	Fashion retail brand #1

systems, where each loop presents distinct challenges that must be addressed to achieve effective scaling.

Performing PTs appeared predominantly at the product level (sub-themes #1–4), both within inner and outer circular loop CBMs. Within the inner-loop CBMs, brands, like Fashion retail brand #6, point out that tensions arise when trying to scale service-oriented CBMs like repair to reduce new product usage while staying economically viable. Garment repairing is expensive, and, without the support of relevant tax exemptive policy instruments, as highlighted by Brändström et al. (2024), results in higher price points compared to buying cheaper fast fashion in linear retail models, thus inhibiting economic viability. In the resale market, growth of second-hand businesses, particularly for workwear/functional garment segments, is hindered by such performing PTs as well, as concerns rise regarding ensuring product safety, ethics, security, and guaranteeing product functionality and performance, as was revealed by multiple actors (e.g., Recycler #1, Integrated sorter #2, and Second-hand charity #4). This is in line with PTs observed by Mittal et al. (2024) in their study of the footwear industry, where hygiene-related challenges impede the promotion of various CBMs, including rental, sequential sharing, and swapping.

In outer-loop CBMs, i.e., recycling, performing PTs largely hinder the market uptake of recycled materials due to several reasons, such as technology limitations, circular product data gaps and higher price points, as shown in Table 2. Extending product life through design for durability by incorporating fiber blends unfolds tensions to fulfill recyclability, as noted by Fashion retail brands #3 and #4. In a similar vein, Mishra et al. (2022) highlights that existing technologies are not capable of efficiently processing all types of textile waste and blends, thereby hindering large-scale recycling operations. These conflicting approaches, with different outcomes, are inevitable as companies transition to CE practices and must adapt their value-creation logic (De Angelis, 2021). Furthermore, the significant investment required for new machinery to reintegrate recycled materials into manufacturing (Jaeger and Upadhyay, 2020) leads to higher processing costs (Brändström et al., 2024), resulting in good quality materials being costly in the market leading to lesser customer acceptance, as emphasized by Fashion retail brand #6. In contrast, scaling use of recycled fibers can be hindered by safety and hygiene issues, as highlighted by Daddi et al. (2019). This can create a vicious cycle between supply and demand, as argued by Reike et al. (2023), thus making brands reluctant to switch gear in accelerating uptake of recycled textiles, which in turn causes recyclers hesitant in investing in large-scale facilities. All these performing class PTs stem from socio-environmental and economic conflicts and cause actors to make trade-offs about what actions to pursue or avoid (Smith and Lewis, 2011). Accordingly, we propose:

Proposition 1. Performing PTs primarily hinder scaling CBMs by making circular products or materials more expensive and less accepted by customers due to cost, quality, and safety concerns, thus limiting CBM market growth.

Organizing PTs are particularly evident in the outer-loop CBMs, specifically within the recycling value chains (sub-themes #5, #7–8, and #11). First, these PTs arise when choosing between recycling technologies (i.e., mechanical and chemical) while scaling the use of recycled materials. According to Sandvik and Stubbs (2019), recycling processes involve inherent trade-offs between fiber quality and characteristics. These trade-offs, in turn, impact scaling operations, as the choice between these methods influences the quality and price of the final product, thus affecting market uptake. Second, these tensions emerge when attempting to scale recycling operation while managing feedstock variability. On the one hand, efficient processing of post-consumer textiles requires sorting infrastructure and optical technology, such as near-infrared (NIR), to accurately identify fiber composition (Boschmeier et al., 2024). The lack of such infrastructure and technology presents significant challenges in managing variable feedstock quality and economies of scale in processing, as noted by Integrated sorters #4 and #5. On the other hand, the lack of credible recycling value chains, as highlighted by Recycler #1, restricts their ability to scale the production of recycled materials and products. The latter, especially in the case of immature chemical recycling supply chains, which is accompanied by ad-hoc sourcing solutions, urges a more structured approach in sorting feedstock, as noted by Recycler #3. Finally, these PTs arise when deciding the right geographic spatiality for organizing full-scale recycling operations. Although local operations offer greater flexibility to meet specific market needs (De Angelis, 2021; Greer et al., 2021), capacity constraints can be the biggest issue to achieve economies of scale for cost-efficient processing of large volumes of materials as pointed out by Integrated sorter #6. However, globalized recycling value chains, while potentially more scalable, incur higher transportation costs and a larger environmental footprint (Sandberg and Pal, 2024). Therefore, maintaining a balance among these factors is essential for effective and sustainable recycling CBMs. All the above-mentioned tensions stem from competing processes and value chain structures and design, which are prime examples of organizing class paradoxes described by Smith and Lewis (2011). Considering the above, we propose:

Proposition 2. Organizing PTs primarily hinder the scaling of recycling-based CBMs due to quality-cost trade-offs in selecting alternative technologies, the lack of availability of credible yet cost-efficient value chains to manage variable feedstock, and the challenge of choosing geographic spatiality to enable a sustainable recycling value chain.

Among inter-class paradoxes, organizing-performing ones were the most prominent, experienced by actors at all levels and across different CBMs, though they were more noticeable at the process and network levels. This suggests that actors in the T&C reverse supply chain encounter tensions from the interplay of “means and ends”, or process and outcome (Smith and Lewis, 2011). These tensions arise when actors strive to balance operational efficiency and effectiveness (sub-themes #6 and #10–12). On the one hand, volume-based competition and the cannibalization of used clothing between competing CBMs, as stated by Fashion retail brand #5 and Integrated sorter #5, hinder the individual scaling of CBMs. On the other hand, the prevalence of new and affordable fast fashion renders waste processing costs uneconomical (Schmutz and Som, 2022), leading to lower-value recovery such as incineration, as pointed out by Integrated sorters #2.

In resale-based CBMs, such inter-class tensions also emerge when balancing global/regional vs. local and centralized vs. decentralized value chains. For example, local supply chains cannot gain economies of scale due to high operational costs and a lack of skilled local workforce for sorting. Seeking a low-cost and skilled workforce on a regional or global scale also incurs high logistics costs for pre-sorting reusable items, as stated by Second-hand charity #1. Furthermore, prioritizing decentralized operations to align with store values results in sorting inefficiencies, in contrast to the cost efficiency and economies of scale achieved through centralized operations, as highlighted by Second-hand charity #4. This is consistent with the study by Khorshidvand et al. (2023) on H&M's supply chain, which found that centralized operations enhance supply chain efficiency and overall profitability. However, while decentralized operations provide social benefits, such as volunteer work opportunities – which are integral to these charities' work philosophy – their inefficiencies in scaling reuse-based CBMs may ultimately result in diverting reusable products to outer-loop CBMs, such as recycling, rather than maximizing their potential for direct reuse. This demonstrates how different value chain configurations can hinder scaling due to costs or skill-related challenges. Since these inter-class organizing-performing tensions significantly obstruct the scaling of sequential loops within inner and outer-loop CBMs, thereby undermining the CE waste hierarchy, we propose:

Proposition 3. Organizing-performing tensions are the most visible common inter-class paradoxes arising between sequential loops of CBMs mainly due to volume-based competition leading to cannibalization, and the high processing costs associated with operationalizing their underlying reverse value chains.

The belonging and learning PTs were less prominent and typically appeared alongside other paradox classes. Specifically, inter-class belonging PTs were noticeable at the stakeholder level (sub-themes #13–15). First, these tensions manifested as a performing-belonging paradox, where identity tensions clashed with sustainability objectives (Smith and Lewis, 2011). This conflict arose from differing values, ownership models, and ideological differences, leading to misaligned socio-economic goals within the CE. For example, Second-hand charity #4 experienced this tension in scaling resale operations. Second, they arose as an organizing-belonging paradox, emerging at the intersection of the individual and the aggregate (ibid.). For instance, cognitive challenges in transitioning to circularity arose when shareholder value and established operational processes were prioritized over circular strategies, as noted by Fashion retail brand #1. These implicit identity tensions, which impede scaling efforts across various CBMs, expand the insights provided by van Bommel (2018). Distinctively, an inter-class organizing-performing-belonging tension emerged when attempting to balance cost-effectiveness with accountability in the value recovery process due to regulatory conundrums. For instance, since designing waste management systems is challenging (Johansson, 2022), the lack of financial motivation, as stated by Second-hand charity #1, may further disengage actors from supporting circularity efforts. This is a unique inter-class PT, which was not explicitly classified by Smith and Lewis (2011).

Inter-class learning PTs are exclusively seen as organizing-learning paradoxes, appearing at both the product and process levels. At the product level, this tension arises from the trade-off between durability and the ability to scale recyclability (sub-theme #1). This tension is driven by limited market uptake due to technological constraints and insufficient data on product and material lifespans, as noted by Integrated sorter #4. At the process level, this inter-class tension (sub-theme #7) emerged due to the high processing costs and knowledge gaps in scaling CBMs and the risk of cannibalizing mainstream CBMs, such as resale, when diverting more volumes for repair and remaking. This finding aligns with Jarzabkowski et al. (2013), who describe learning as a complex, multi-layered concept that can intersect with other paradox classes. Therefore, similar to belonging PTs, learning PTs can inherently arise, as achieving long-term sustainability relies on systemic innovation and maintaining a delicate balance between exploration and exploitation (Hahn et al., 2018; Schad et al., 2016). According to Smith and Lewis (2011), inherent PTs become salient under conditions of plurality,

where diverse stakeholder expectations alongside human resource constraints create complexities, as observed in the case of Second-hand charity #5. In light of the above, we propose:

Proposition 4. Learning and belonging paradoxes are least explicit and arise alongside other paradoxes. Inter-class belonging PTs hinder CBM scaling by creating ideological, cognitive, or accountability challenges, as these stem from conflicting stakeholder interests and the impact of differing policy mechanisms. In contrast, inter-class learning PTs hinder scaling by creating gaps in knowledge, technology, and data within CBMs.

While outlining the above propositions, we acknowledge that our respondent pool was limited mainly to top managers, CEOs, and key decision-makers. Their broad perspectives and overarching insight into company operations allowed us to identify a wide range of PTs, particularly those related to organizing and performing classes. However, to gain a deeper understanding of intra-organizational PTs that might impede scaling operations, it is essential to include employees from different organizational levels. This is especially pertinent for exploring belonging and learning paradoxes. As Smith and Lewis (2011) emphasize, tensions arising at different tiers of the organizational hierarchy can trigger new challenges across levels. For example, exploration-exploitation tensions can arise when top management advocates for innovation and long-term growth, while employees concentrate on optimizing current processes and maintaining operational efficiency (ibid.).

6. Conclusion

This study focused on exploring PTs that hinder scaling T&C CBMs. Our findings reveal that these PTs emerge across four levels – product, process, network, and stakeholder – while aligning with four paradox classifications: organizing, performing, belonging, and learning. They were identified among both upstream and downstream actors, affecting their scaling efforts within both inner- and outer-loop CBMs. Based on these findings, we present four key propositions that illustrate how scaling challenges arise at different levels and the underlying reasons behind them. Understanding these PTs and the factors that impede scaling is essential for developing strategies that balance competing priorities and enable sustainable growth. These insights contribute to a deeper understanding of the structural and strategic challenges associated with scaling CBMs.

The findings of this study advance theoretical knowledge in two key ways. First, our empirical insights extend paradox theory by conceptualizing PTs within the CE in the T&C industry. By identifying and categorizing these PTs, the study illustrates how intra- and inter-class paradoxes are observed within circular systems, shedding light on the trade-offs and competing demands that arise when scaling CBMs. This contributes to a deeper understanding of organizational and industry-wide paradoxes. Second, this research addresses gaps in CBM scaling literature by showing that PTs extend beyond operational and technical aspects to include stakeholder relationships, regulatory environments, and market structures. By integrating paradox theory with CBM literature, the study provides a foundation to explore strategies for managing tensions, fostering collaboration, and overcoming scaling barriers.

This study also provides valuable insights for both industry professionals and policymakers by presenting a structured framework across four levels – product, process, network, and stakeholder – and identifying 15 sub-themes of PTs that impede the scaling of CBMs in the T&C sector. For businesses, these findings can guide strategic decision-making by pinpointing common trade-offs (e.g., between economic viability and environmental goals at the product level), enabling firms to create more resilient CBMs. For example, companies can proactively address PTs by aligning product design with scalable end-of-life solutions such as modular repair or localized recycling operations. For policymakers, the study highlights the necessity of designing multi-level policy interventions that tackle both structural and behavioral barriers. This includes supporting infrastructure development for high-quality recycling, standardizing product traceability, and promoting data-sharing platforms to reduce information asymmetries – each of which directly addresses organizing and learning paradoxes. Additionally, governments can enhance EPR schemes by offering financial incentives (e.g., tax breaks, subsidies, or innovation grants) to businesses that invest in circular technologies or adopt reuse-repair models. This approach not only encourages actors to take a more active role in circular systems but also helps stabilize the system itself by minimizing the emergence of new PTs, especially those caused by fragmented responsibilities or misaligned stakeholder identities. Effective waste governance mechanisms can further help manage variable material flows and prevent systemic tensions resulting from poorly aligned local-global recycling infrastructures.

This work paves the way for future research in several areas. First, replicating similar studies in various geographical regions and involving a wider range of actors would be beneficial, as it would help evaluate and refine the propositions developed in this study, thereby enhancing their accuracy. Moreover, a longitudinal approach could provide deeper insights into how these tensions evolve over time. Lastly, future research should investigate how organizational capabilities, and adaptive strategies can effectively manage these PTs, offering guidance for practitioners and mediators aiming to facilitate more scalable and resilient CBMs.

CRediT authorship contribution statement

Mohammadreza Dehghannejad: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Rudrajeet Pal:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Kanchana Dissanayake:** Writing – review & editing, Supervision, Methodology, Formal analysis, Data curation. **Erik Sandberg:** Writing – review & editing, Validation, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary Data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.spc.2025.04.022.

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